

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

Department of Electrical Engineering - ICDT

PROJECT REPORT

Arduino-Based Digital Thermometer Using a PT100 Sensor

Submitted by:

Rudra Vishweshwar (EE26BTECH11225)

Saksham Sawant (EE26BTECH11226)

Ganesh Balotiya (EE26BTECH11205)

Submitted To:

Prof Dr.GVV Sharma

Teaching Assistants:

Vivek Kumar, Subhodeep Chakraborty , Jnanesh

Date of Submission:

Oct 16, 2026

Table of Contents

1 Introduction 1

2 Project Objectives 1

3 Component Overview 2

3.1 List of Materials 2

3.2 Brief Overview of Components 3

4 Working Principle 4

5 Testing and Temperature Measurements 5

6 Circuit Assembly on Perf-Board 6

6.1 Component Placement 6

6.2 Soldering the Components 6

7 Code 8

8 Case Designing & 3D Printing 9

8.1 Final Case Design 9

9 Problems Faced 10

10 Completion and Conclusion 11

10.1 Functional Thermometer 11

10.2 Conclusion 11

List of Tables

1 List of Components 3

List of Figures

1 Components for thermometer 2
2 Thermometer testing 5
3 Soldered Top 7
4 Soldered Bottom 7
5 Final Case Design 9

1 Introduction

This project is about building a digital thermometer that can measure temperature. It involves connecting different electronic components with an Arduino UNO to make a working thermometer. Through this project, we aim to learn how to build electronic circuits and use an Arduino in a practical application.

2 Project Objectives

- To build a digital thermometer using an Arduino UNO and a PT100 temperature sensor.
- To display the measured temperature on a digital display.
- To learn how to connect electronic components and program an Arduino UNO.
- To assemble the circuit and solder neatly on a perfboard.
- To design and 3D print a case for the Thermometer.
- To test the thermometer and verify its output temperature.
- To gain hands-on experience in circuit assembly, with electronics, soldering, and 3D printing.

3 Component Overview

The thermometer is implemented using discrete hardware components distributed for the laboratory project. This section details the list of materials used in the physical assembly.

3.1 List of Materials

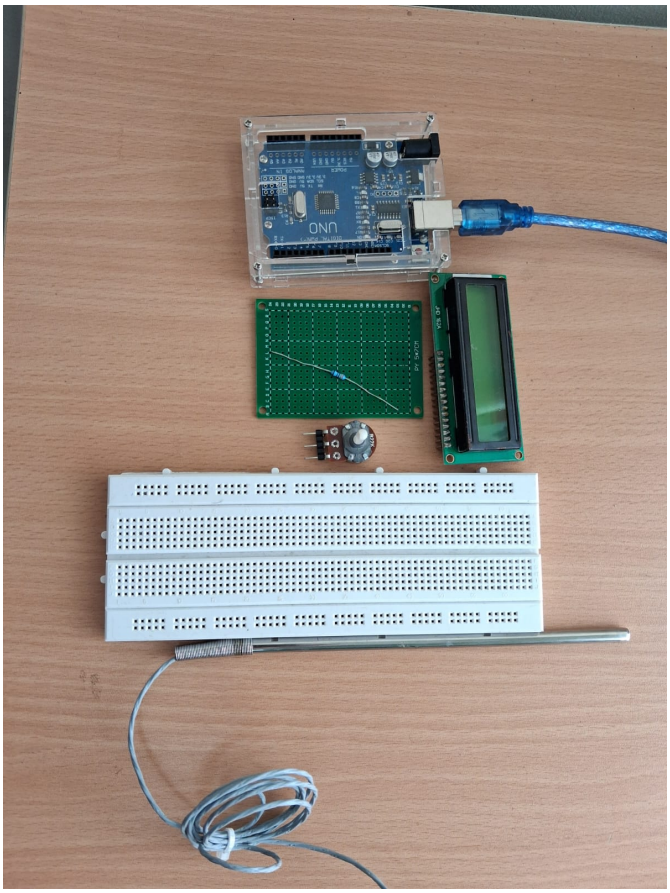


Figure 1: Components for thermometer

Table 1 outlines the specific parts received and integrated into the perfboard setup.

Table 1: List of Components

S.No.	Component Name	Quantity
1	16×2 LCD Display Module	1
2	Arduino UNO	1
3	pt100 Temp sensor	1
4	Perf Board	1
5	Resistor (Ω)	1
6	Potentiometer	1
7	Bread Board	1
8	Connecting Wires and Solder	as needed

3.2 Brief Overview of Components

- **`Arduino UNO:`** It is a microcontroller board that is used to control the thermometer and process the temperature readings.
- **`Potentiometer:`** It is an adjustable resistor that can be used to control the contrast of the LCD display.
- **`PT100 Temperature Sensor:`** It is a temperature sensor whose electrical resistance changes with temperature. It is used to measure the temperature.
- **`LCD Display:`** The LCD display is used to show the measured temperature in a readable format.

4 Working Principle

The thermometer uses a PT100 sensor to measure temperature, an Arduino UNO to process the readings, and an LCD to display the result.

- The PT100 is a platinum-based temperature sensor whose electrical resistance changes with temperature. At 0°C, its resistance is 100Ω. Its resistance increases as the temperature rises and decreases as the temperature falls.
- The measurement circuit converts the change in PT100 resistance into a corresponding voltage that can be measured by the Arduino UNO.
- The Arduino reads the voltage through an analog input pin. Its analog-to-digital converter (ADC) converts the voltage into a digital value for processing.
- The Arduino program uses the measured value and the relationship between the PT100's resistance and temperature to calculate the corresponding temperature.
- The calculated temperature is sent to the LCD display, allowing the user to observe the reading as the temperature changes.
- The voltage regulator provides a suitable supply voltage to the circuit, depending on its connection in the design.

In summary, we can say the working principle as:

Temperature Change → PT100 Resistance Change → Voltage Signal Generation → Analog-to-Digital Conversion → Arduino Temperature Calculation → LCD Display of Temperature.

5 Testing and Temperature Measurements

- The thermometer circuit was first assembled on a breadboard to test the connections and check whether the components were working as expected. This allowed us to make changes to the circuit easily before making permanent connections.

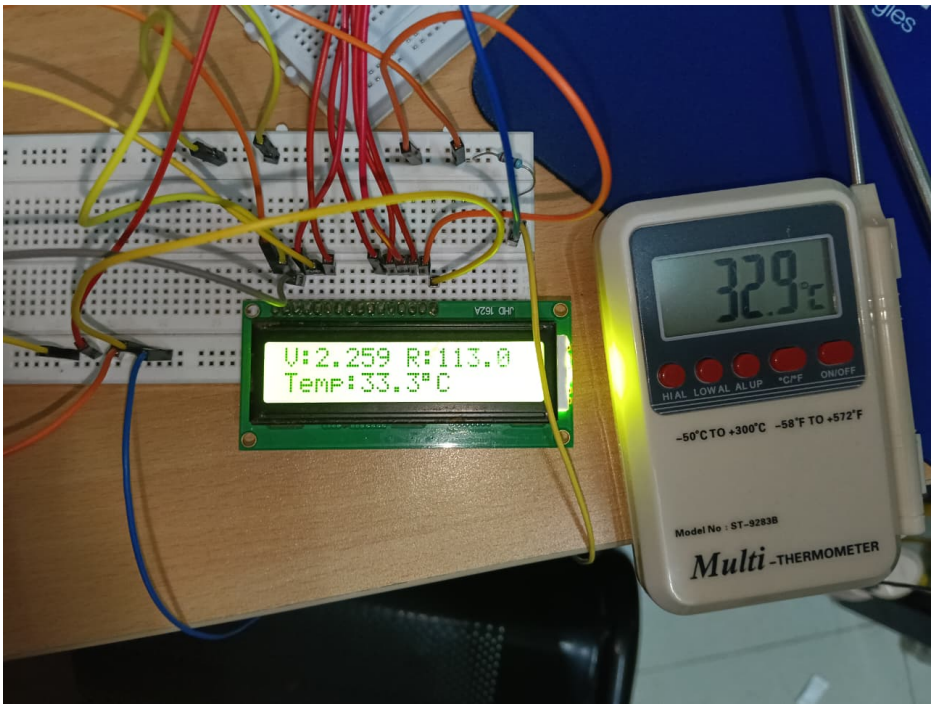


Figure 2: Thermometer testing

- During testing, the thermometer displayed a temperature of 33.3°C, while a separate thermometer showed 32.9°C under the same conditions.This indicated that the thermometer gave a reading close to the reference thermometer during the initial test.

- After testing the circuit on the breadboard, the readings helped us confirm that the circuit was working as expected.We proceeded to assemble and test it on a perfboard to check its operation with the components mounted permanently.

6 Circuit Assembly on Perf-Board

6.1 Component Placement

- The components were placed on the perfboard, according to the circuit diagram in fig 2,keeping the connections as compact and organized as possible and the assembled perfboard looks like the Fig 5.
- The PT100 sensor and other circuit components were connected to the Arduino UNO using jumper wires.

6.2 Soldering the Components

The components were soldered onto the perfboard according to the circuit connections. During soldering, care was taken to make the joints secure while avoiding excessive solder and unnecessary wire ends. The connections were kept clean and organized to give the circuit a neat overall appearance. Especially since the pins of the lcd were so close together, soldering has becoming challenging.

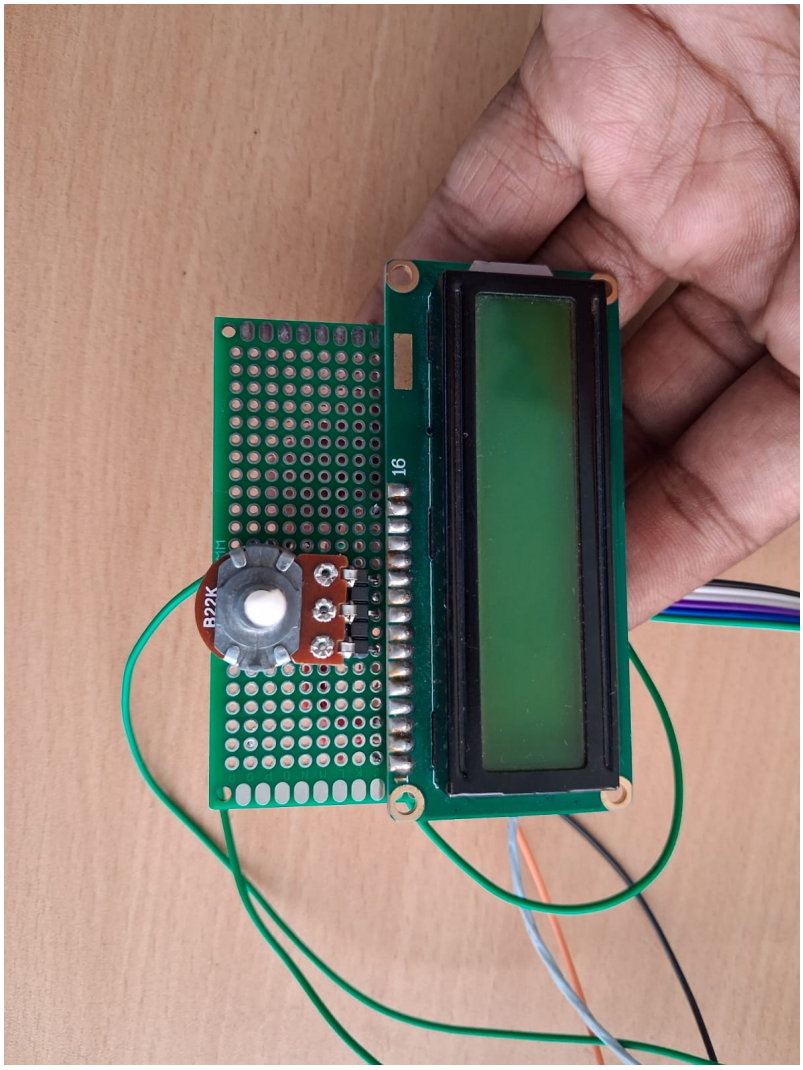


Figure 3: Soldered Top

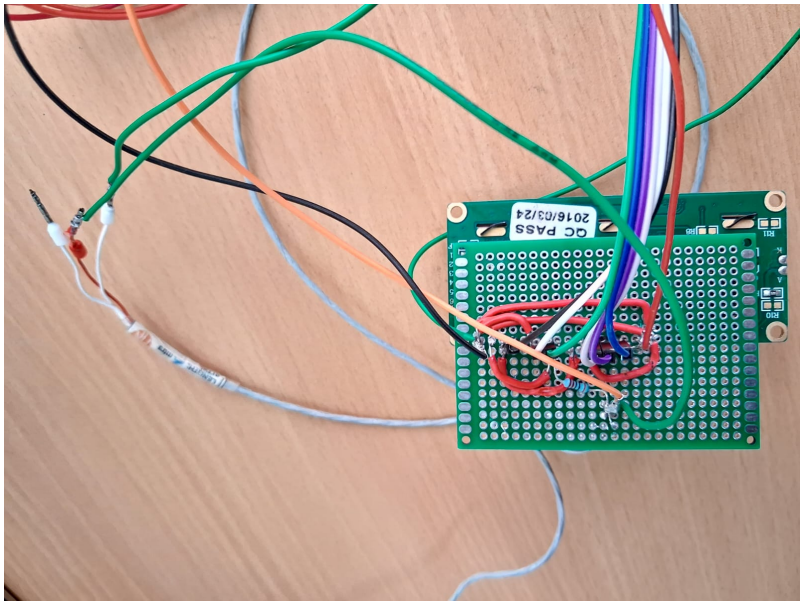


Figure 4: Soldered Bottom

7 Code

Use of Least-Squares Method

In our PT100 thermometer project, we considered the least-squares method for estimating temperature from the resistance measured by our Arduino code.

The least-squares method is a mathematical technique used to find the best-fitting relationship between a set of data points. In our project, it can be used to establish a relationship between the resistance of the PT100 sensor and the corresponding temperature.

The PT100 sensor's resistance changes with temperature. By using known resistance and temperature values, a mathematical relationship can be obtained to estimate temperature from the measured resistance. The least-squares method determines the line that best fits the calibration data by minimizing the sum of the squared differences between the actual and estimated values.

Our current Arduino code calculates temperature using a linear approximation of the PT100 resistance-temperature relationship. Applying least-squares fitting to calibration data could be a future improvement to refine this relationship.

8 Case Designing & 3D Printing

This was easily the most frustrating part of the project. Getting the measurements, component clearances, and overall shape right took several attempts.

8.1 Final Case Design

The following image shows our final case design, made using OpenSCAD.

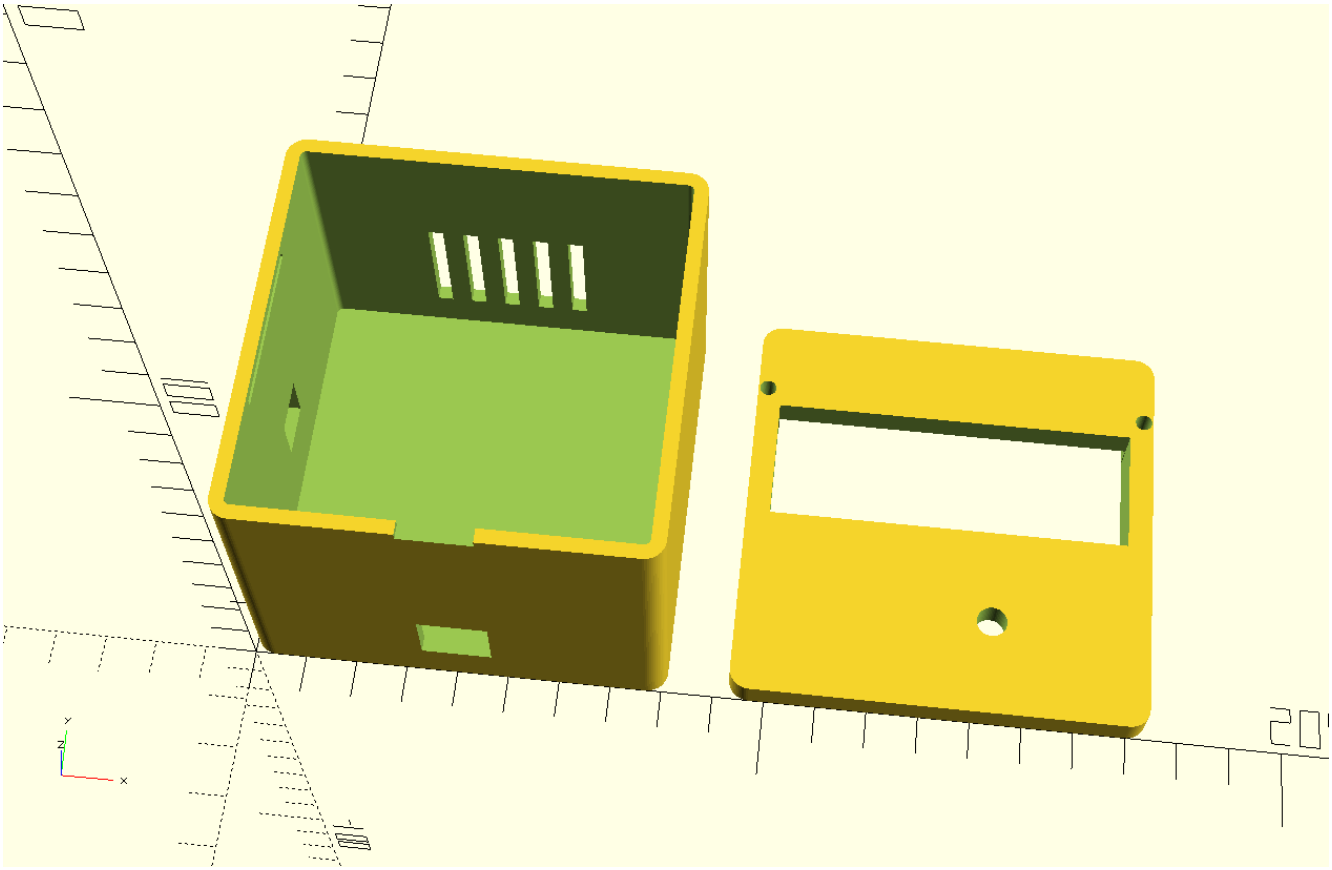


Figure 5: Final Case Design

Particular attention was given to the dimensions of the enclosure so that the thermometer components could fit inside comfortably. The dimensions were adjusted several times as the model developed.We also didn't forget to account for the component clearance space.

9 Problems Faced

Compared to the prev charger project, this project was relatively difficult and involved relatively more difficulties. These challenges helped us identify areas for improvement and make the necessary changes to the final design.

- The main challenge we faced during this project was the limited time available to complete it. Although we were given one week, exams, quizzes, and other academic work kept us busy throughout the week. As a result, we could dedicate only around two days to the project. We had to manage our time carefully to complete the circuit assembly, testing, and 3D case design within the given time.
- Ensuring that solder joints were secure and that nearby connections didn't accidentally touch each other. Since many joints are close together, it was a bit tricky.
- The readings of the thermometer were fluctuating heavily. We used AI to check whats wrong, cross check and modify the code for it.
- At the last moments, some jumper wires broke off, leaving their tips stuck in the arduino itself. We had to take care of it too which was troublesome.

10 Completion and Conclusion

10.1 Functional Thermometer

After completing the circuit, we tested the final thermometer with a water kettel and used a industrial Multi Thermometer, for refference. The Thermometer was able to show the temp readings successfully, showing that the final circuit was working properly.

10.2 Conclusion

Overall, we were able to successfully complete our thermometer and its enclosure. We started with a basic idea and slowly worked on it until we had a working product.

The best part of the project was seeing the thermometer actually work, even though we were getting a bit of fluctuations and errors. It was a good experience for us because we got to work on both the circuit and the physical design of the product. Overall, the project helped us understand how an idea can be turned into a working product through practical work and trying things out ourselves.